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DBbun Invisible Illness Burden Simulator: quality assurance log

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How quality was checked

Check	Who or what did it	What it found	What changed
Automated tests	Python test suite (unittest)	233 automated tests pass. Five tests for a retired two-model workflow are intentionally skipped; they are not failures.	Every defect fixed gets a regression test.
Own five-part audit	Author-directed audit using Claude Code with Claude Sonnet 5.5; read-only first, reports kept	Accuracy, function, federal-resource, interface and wording problems.	Fixed in batches, then re-tested.
AI model critiques of earlier versions of every simulator PDF (25) and comparison PDF (13)	Claude Sonnet 5.5, Grok 4.5, Gemini 3.5 (Google), GPT (OpenAI; exact build not exposed) and GPT-5 Codex, given one rubric-based prompt	Reviewers rated the comparison PDFs 3 to 4 of 5 and raised the same themes across simulators: price-year and currency labels, non-U.S. data compared with U.S. figures, text that broke the PDF layout, over-confident model wording.	Fixed once in the code and prompts, not simulator by simulator. All 25 simulators were then regenerated and the 13 comparisons rebuilt (about $21 in model calls).
AI code review	GPT-5 Codex; report only, it changed no code	Ten findings.	Nine fixed with regression tests; one documented (the browser token is a single-user limit).
Federal cross-checks	Code, not the AI model	Twelve of 25 simulators have at least one computed federal operation; ten compare a figure with federal data. Other figures may receive context only.	The tool distinguishes computed comparisons from context-only figures and states limits on comparability.
FHIR export	Official HL7 FHIR validator, R4 4.0.1, terminology checking off. The check covered structure only; codes were not looked up in an online code library, because they are DBbun's own labels, which no standard library contains.	26 files (25 simulators and one reuse demonstration; the second reuse demonstration was not validated): 0 errors. 40 to 74 best-practice warnings each: no patient, no performer or time, no narrative text, DBbun-defined codes.	First run gave 28 errors per file (undefined DBbun extensions); fixed by shipping their definitions with each bundle.
Public demo	Rebuilt from the same code and opened live	The 25 library simulators and 13 comparisons load; FAQ and downloads match the local tool.	Rebuilt with the audio overviews, simulator numbering, the feedback log and the accuracy fixes.
Generated Python check	check_generated_python.py, run locally October 10, 2026	The downloadable Python file in each of the 25 library bundles reproduced the app's numbers: 593 results (107 at default assumptions, 486 across 115 named scenarios) and 12 forecast series (only 5 simulators have a forecast). All 605 were identical.	New simulators' files are not run automatically.
Second AI review of the whole package	GPT (OpenAI; exact build not exposed), October 10, 2026; report only	Excel exports hold saved values and formula text, with no working formulas or recalculation; one comparison said figures were "not provided" while citing them; some cost ranges and PDF text were stale.	Wording corrected, that comparison regenerated with a code check, stale text fixed.

How reviewer advice was handled

I fixed only problems that were real and systematic, checking each against the paper or the data first. Wording that claimed more than the evidence showed was not used. The reviewers are tools, not authorities: I decided what to adopt.

What this log does not show

- The AI critiques covered earlier versions. The regenerated outputs received automated checks and my own spot checks, without a second full AI critique.
- No independent person has yet completed a documented, uncoached task in the public demo. Two outside readers (a neighborhood-group member and a person with lived experience of Lyme disease and a research background) gave written feedback without doing the tasks: the first screen was too dense, the page text-heavy and my time estimate too short. I added a skip-the-grid note, a collapsed coverage map, one-line study summaries, a Lyme / Long COVID filter, a "what this is and is not" banner and FAQ pictures, and shortened my request.
- The 41 spoken overviews (25 simulator, 13 comparison, 3 about the tool) use a synthetic voice. Of the 38 simulator and comparison scripts, three received an additional paragraph-by-paragraph review using Claude Code; the other 35 received automated rule checks only. The three tool overviews were written with Claude Code and revised to my wording rules. The FHIR export is a prototype: not clinical data, not profiled to US Core or USCDI+, not certified; terminology checking was off.
- The two demonstrations on other conditions (a paper on post-surgical chronic pain and an emergency-care cost paper) used the general pipeline only. The cost paper's dollar amounts were restated in current prices by the price-index check; neither has curated federal evidence for its condition or outside validation.

AI tools and model IDs

Inside the product: Claude Sonnet 5 (claude-sonnet-5, Anthropic) reads the papers and writes each specification and its wording; code checks the generated specification against defined rules and performs the simulator arithmetic, and those checks do not establish that extraction or tagging is correct. Building and testing: Claude Code (Anthropic) with Claude Sonnet 5.5 (claude-sonnet-5-5). Audio overviews: scripts written with Claude Code; voice by Kokoro-82M, an open-source text-to-speech model run locally. Reviews: Claude Sonnet 5.5, Grok 4.5 (xAI), Gemini 3.5 (Google), GPT and GPT-5 Codex (OpenAI; exact builds not exposed). I designed and directed the simulator. AI tools wrote code and text under my continuous review, feedback and modification; people who gave feedback are not authors, co-developers or sponsors. Interviewees, forum respondents and outside testers are described by role only.